Topic Test 3 · Paper 1

Question 1

Compute the integral

$$\int_0^1 \frac{x-4}{\sqrt{x}+2}\,dx.$$

A $-\frac{4}{3}$

B -3

C -1

D $\frac{2}{3}$

E 1

Topic Test 3 · Paper 1

Question 2

Given

$$f(x) = \left(9x^2 + 12 + \frac{4}{x^2}\right)^{1/2}$$

and

$$\frac{d^n f}{dx^n}(2) = -\frac{3}{4},$$

find n.

A 1

B 2

C 3

D 4

E 5

Topic Test 3 · Paper 1

Question 3

For $p > 0$, find the area enclosed by the curves

$$y = px^2 \qquad \text{and} \qquad x = py^2.$$

A $\frac{1}{3p}$

B $3p$

C $\frac{1}{3p^2}$

D $\frac{1}{2}p^2$

E $\frac{1}{6p^2}$

Topic Test 3 · Paper 1

Question 4

What is the complete set of values for which

$$\frac{x^2 + 2x}{\sqrt{x^3}}$$

is increasing?

A $x > 3$

B $x < 2$

C $2 < x < 4$

D $x > 2$

E $0 < x < 2$

Topic Test 3 · Paper 1

Question 5

A point P lies on the part of the parabola $y = x^2$ in the first quadrant. What is the shortest possible distance from P to the point $(0, 3)$?

A $\frac{\sqrt{11}}{2}$

B $\frac{3}{2}$

C $\frac{\sqrt{5}}{2}$

D $\frac{11}{4}$

E $\sqrt{3}$

Topic Test 3 · Paper 1

Question 6

Given that

$$\frac{dV}{dt} = (1+t)^4$$

and $V(1) = 5$, what is $V(2)$?

A $\frac{174}{5}$

B $\frac{236}{5}$

C $\frac{112}{3}$

D $\frac{89}{4}$

E $\frac{17}{32}$

Topic Test 3 · Paper 1

Question 7

Let

$$y = \left(\frac{1}{3}k^3 + 1\right)x^3 - (2k^2 + 3)x^2 + (31 - 25k)x + 43.$$

The curve has a turning point that is a maximum at $x = 1$, precisely for:

A $-4 < k < 7$

B $k = 1$ or 7

C $k > 7$

D $k = -4$ or 7

E $k < 1$

F $k = -4$ or 1

G $k < -4$

H $k = -4, 1$ or 7

Topic Test 3 · Paper 1

Question 8

Which of the following integrals has the largest value?

A $\displaystyle\int_{0}^{\pi/3} \cos\left(\frac{\pi}{2} - x\right) dx$

B $\displaystyle\int_{\pi/3}^{2\pi/3} \cos^2 x \, dx$

C $\displaystyle\int_{2\pi/3}^{\pi} -\sin\left(\frac{\pi}{2} - x\right) dx$

D $\displaystyle\int_{0}^{\pi} \sin\left(\frac{\pi}{2} - x\right) dx$

Topic Test 3 · Paper 1

Question 9

Find the value of

$$\int_{-1}^{1} \frac{x(x^2-2)^2}{x^2+1}\,dx.$$

A 0

B 1

C 2

D 3

E $\frac{3}{7}$

F 12

Topic Test 3 · Paper 1

Question 10

A curve has equation

$$y = (2a - x)(2a + x),$$

and a second curve has equation

$$y = (2a - x)(3a + x),$$

where $a \neq 0$. The distance between the maximum points of the two curves is $\sqrt{10}/3$. Find the possible values of a.

A 0

B $\pm\frac{2}{3}$

C $\pm\frac{1}{2}$

D $\pm\frac{\sqrt{10}}{3}$

E ± 1

F $\pm\frac{2\sqrt{5}}{5}$

Topic Test 3 · Paper 1

Question 11

A function f satisfies

$$\int_0^4 f(x)\,dx = 10, \qquad \int_{-1}^4 f(x)\,dx = 13, \qquad \int_{-4}^3 f(-x)\,dx = 2.$$

Find

$$\int_{-3}^0 f(x)\,dx.$$

A 2

B 3

C 5

D -8

E 11

F 12

Topic Test 3 · Paper 1

Question 12

A pump's discharge rate, in litres per minute, is measured at times $t = 0, 2, 4$ minutes. The readings are $3, 9, 5$, respectively. A fourth reading is found at $t = 3$. Using all four readings, the trapezium estimate of the total discharge is 28.5 litres. What rate was measured at $t = 3$?

A $\dfrac{19}{2}$

B $\dfrac{13}{2}$

C 7

D 8

E $\dfrac{17}{2}$

F 10

Topic Test 3 · Paper 1

Question 13

Find the complete set of values of k for which

$$2x^3 - 3x^2 - 36x + k = 0$$

has three distinct real solutions.

A $k < 44$

B $k < 81$

C $-81 < k < 44$

D $-44 < k < 81$

E $k > 44$

F $k > 81$

Topic Test 3 · Paper 1

Question 14

A function y satisfies

$$\frac{dy}{dx} = 7 + \frac{3}{x}, \qquad x > 0.$$

Its general solution is written as

$$y = 7x + 3\ln x + C.$$

Given that $y = 5$ when $x = 3$, find C.

A $16 + 3\ln 3$

B $-16 - 3\ln 3$

C $-2 - 3\ln 3$

D $-16 + 3\ln 3$

E $21 + 3\ln 3$

F 5

Topic Test 3 · Paper 1

Question 15

Find the value of

$$\int_0^2 |x-1|\left(3\sqrt{x}-x\sqrt{x}\right)dx.$$

A $\frac{48-4\sqrt{2}}{35}$

B $\frac{4\sqrt{2}}{35}$

C $\frac{24}{35}$

D $\frac{48+4\sqrt{2}}{35}$

E $\frac{24+4\sqrt{2}}{35}$

F $\frac{24-4\sqrt{2}}{35}$

Topic Test 3 · Paper 1

Question 16

For a function f,

$$\left(\int_0^2 f(x)\,dx\right)\left(\int_0^1 f(x)\,dx + 4\right) = 24.$$

Also, $f(x+1) = f(1-x)$. Which of the following could be the value of

$$\int_1^2 f(x)\,dx?$$

I. -6
II. 8
III. 2

A None

B I only

C II only

D III only

E I & II only

F I & III only

G II & III only

H All three

Topic Test 3 · Paper 1

Question 17

Given that

$$\frac{dy}{dx} = \frac{3}{2}\left(\frac{x^{7/3} - x^{5/3}}{x^{1/3} - 1} + \frac{x^{7/3} - x^{5/3}}{x^{1/3} + 1}\right), \qquad x > 1,$$

and $y = 4$ when $x = 2$, find y when $x = 5$.

A 121

B 117

C 125

D 113

E 129

F 81

G $\frac{243}{2}$

Topic Test 3 · Paper 1

Question 18

Given that

$$\int_0^4 (x^3 + ax^2 + bx + 1)dx = 0,$$

where $a \leq 0$ and $b \leq 0$, find the maximum value of

$$\int_0^3 (x^3 + ax^2 + bx + 1)dx.$$

A -15

B 15

C 100

D $-\frac{87}{16}$

E $\frac{123}{2}$

F $-\frac{123}{2}$

Topic Test 3 · Paper 1

Question 19

Let

$$f(x) = \frac{\left(x^2 - \frac{1}{x}\right)^2}{x^{1/3}}.$$

It is also given that

$$f'(x) = \frac{x^{2/3}}{3}(ax^p + bx^q + cx^r).$$

Find $abc + pqr$.

A 388

B $\frac{520}{3}$

C 288

D 316

E 300

F -80

Topic Test 3 · Paper 1

Question 20

Consider

$$g(x) = \int_0^x (t - p) f(t)\, dt, \qquad 0 \le p \le 7.$$

Find

$$\int_0^7 |x - p| f(x)\, dx.$$

A $g(7) - g(0)$

B $g(7) - g(p)$

C $g(0) - p\, g(7)$

D $2g(0) - g(7) + g(p)$

E $g(7) + g(0) - 2g(p)$

F $g(p) + g(7) - 2g(0)$